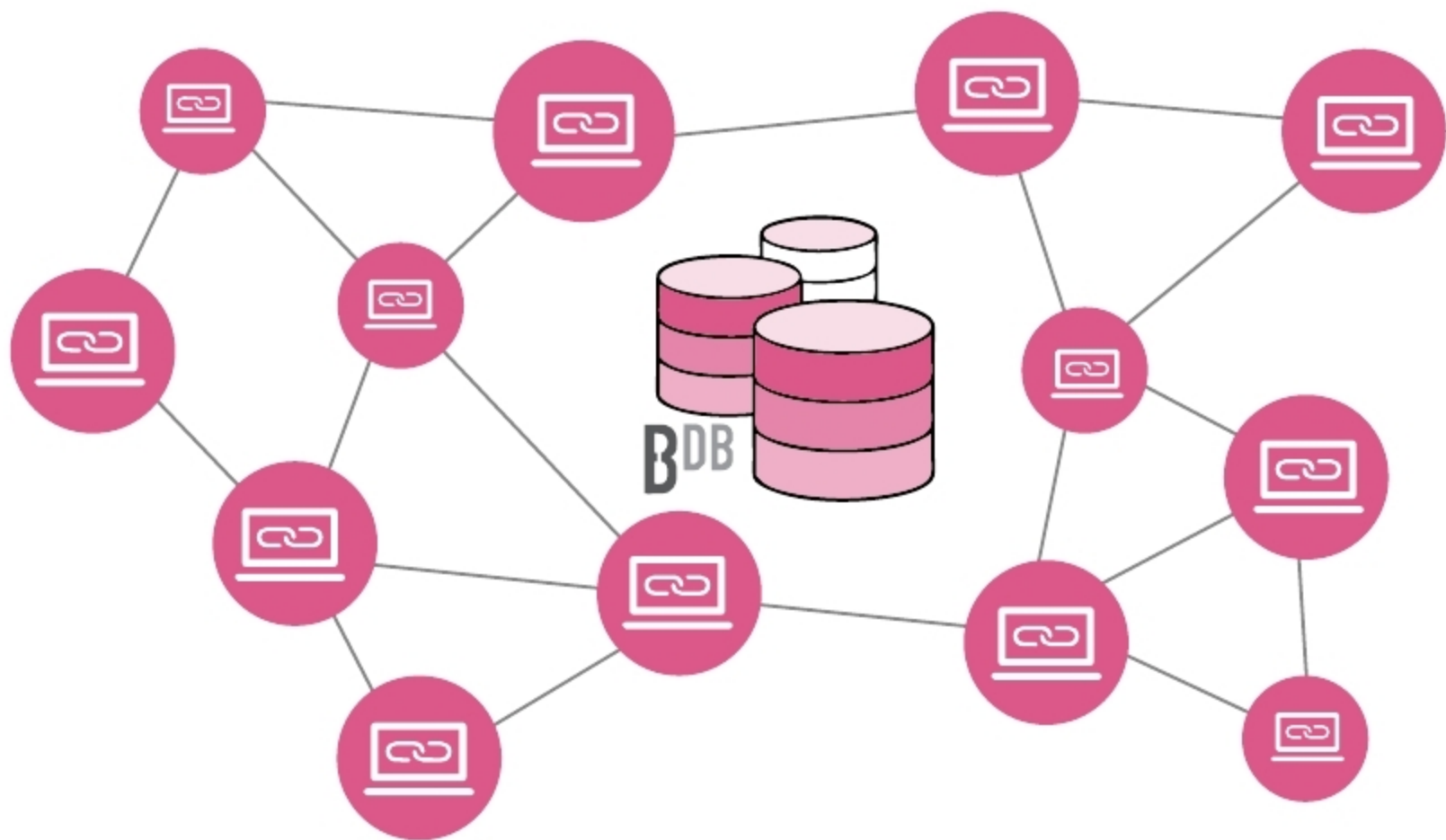


An Introduction to BigchainDB, a Popular Blockchain Database



Blockchain technology provides decentralised digital public records of transactions that are secure, anonymous, tamperproof and unchangeable. It allows transactions without the need for any centralised authority. However, blockchain technology requires a powerful database, and this is where BigchainDB comes in. This article explores all aspects of it.

BigchainDB is a scalable database that allows developers and enterprises to deploy applications with blockchain characteristics. It combines the best characteristics from blockchain technology such as decentralisation, Byzantine fault tolerance, immutability, owner-controlled assets as well as high transaction rates, low latency and scalability from the distributed database domain. BigchainDB uses Tendermint to obtain the features of blockchain

technology and an open source database called MongoDB for storage purposes. Currently, its stable software release version is BigchainDB 2.0.

BigchainDB = Tendermint + MongoDB

Key features and real-time use cases

BigchainDB is widely used in production environments in various real-time applications such as data governance, identity management,

supply chains, financial ecosystems and IoT. The key characteristics of BigchainDB are highlighted in Table 1 and described below.

- **Full decentralisation:** No single entity controls the network. Every node has its own local-stored list of the public keys of other consortium members.
- **Byzantine fault tolerance:** Up to one-third of the nodes in the network can experience arbitrary faults and the rest of the network

